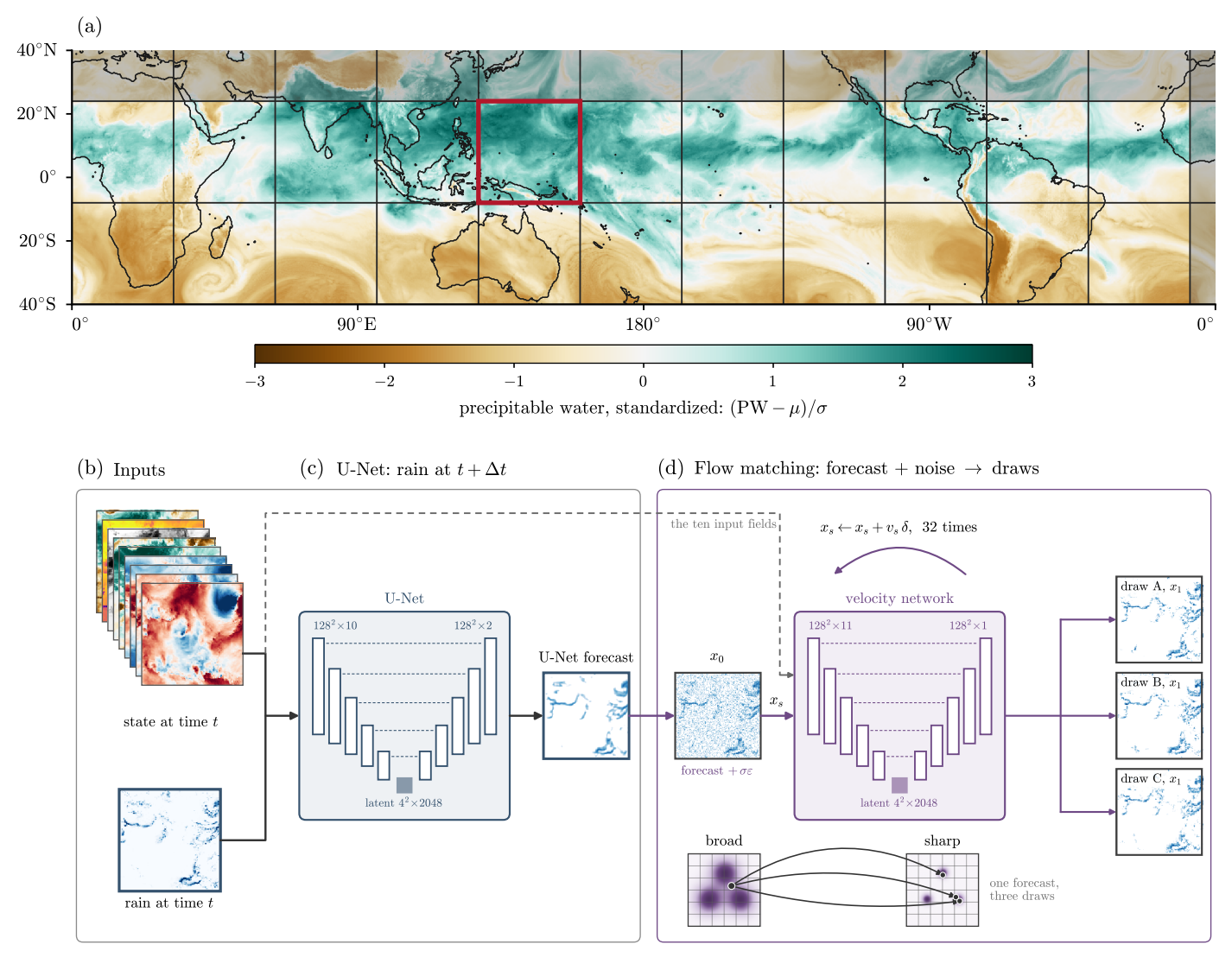

(a)
40°N
20°N
0°
20°S
40°S
0°
90°E
180°
90°W
0°
−3
−2
−1
0
1
2
3
precipitable water, standardized: (PW − μ)/σ
(b) Inputs
(c) U-Net: rain at t + Δt
(d) Flow matching: forecast + noise → draws
state at time t
rain at time t
U-Net
128²×10
128²×2
latent 4²×2048
U-Net forecast
the ten input fields
x_s ← x_s + v_s δ, 32 times
velocity network
128²×11
128²×1
latent 4²×2048
x_0
x_s
forecast + σε
draw A, x_1
draw B, x_1
draw C, x_1
broad
sharp
one forecast,
three draws